FPGA-based mix-precision Monte Carlo using high level synthesis tools

EGH490-1 Project Proposal

September 4, 2026

Author:

Luca Lubrano Lavadera

Under the supervision of:

Dr Jasmine Banks

Dr David J. Warne

1 General Objective

The primary objective of this project is to explore how reconfigurable computing can be used to reduce the total power consumption of heterogeneous computing environment, such as data centres. We will develop practical implementations of Monte Carlo algorithms to benchmark against theoretical techniques that have previously been explored in the literature. To do this, we will perform a case study utilising field programmable gate arrays (FPGAs) and high-level synthesis tools to understand the feasibility of adopting and utilising reconfigurable devices as accelerator hardware in these heterogeneous frameworks. add to — should i be talking about mixed-precision methods here?

1.1 Research Questions

1. To what extent can implementations of Monte Carlo on reconfigurable computing hardware improve the power efficiency of algorithms compared to graphical processing units?
2. To what extent can high-level synthesis tools and portable code frameworks be used to reduce the development time of algorithms in heterogeneous computing environments¿

2 Key Finding from the Literature

todo after literature is finalised

3 Stakeholders & Resources

intro para

3.1 Stakeholders

todo

3.2 Resources

To be able to complete this project there are several key resources that need to be secured. The most critical of these being access to an FPGA on the QUT high performance computer (HPC). This involves the coordination of appropriate software installation and hardware access permissions with Hamish Macintosh, DevOps Team Lead with HPC.

Item	Description	Cost
AMD U55C FPGA	todo	$0.00
Intel CPU	todo	$0.00
NVIDIA GPU	todo	$0.00
Xilinx Vivado Vitis	todo	TBD

Table 1: todo

4 Project Methodology

4.1 Methods

4.2 Results

5 Deliverables

The expected outcome of this project will be the development and verification of algorithms using high-level synthesis tools for FPGA devices. The main contribution of this work will be benchmarking results and analysis in how FPGAs and reconfigurable computing architectures can be implemented in heterogeneous computing environments to reduce power consumption for compute intensive tasks. This will include a detailed report on the usability of, and development time for complex algorithms using, these HLS tools to better understand the feasibility of adopting FPGAs into scientific computing.

Deliverable	Stakeholder	Due Date
EGH490-1 Project Proposal Draft	Supervisors	21/08/26
FPGA Simulation Example Simulation	Supervisors	04/09/26
EGH490-1 Project Proposal	Supervisors	04/09/26
Preliminary Simulation Findings	Supervisors	11/09/26
Geometric Brownian Motion Simulation	Supervisors	02/10/26
EGH490-1 Progress Report Draft	Supervisors	09/10/26
EGH490-1 Progress Report	Supervisors	23/10/26

Table 2: Deliverable outcomes of EGH490-1

Deliverable	Stakeholder	Due Date
EGH490-1 Project Proposal Draft	Supervisors	21/08/26
FPGA Simulation Example Simulation	Supervisors	04/09/26
EGH490-1 Project Proposal	Supervisors	04/09/26
Preliminary Simulation Findings	Supervisors	11/09/26
Geometric Brownian Motion Simulation	Supervisors	02/10/26
EGH490-1 Progress Report Draft	Supervisors	09/10/26
EGH490-1 Progress Report	Supervisors	23/10/26

Table 3: TODO: Deliverable outcomes of EGH490-2

6 Risks, Requirements & Constraints

6.1 Risks

Since this project has key dependencies that must be met in order to progress, specifically access and functionality of the FPGA on the QUT HPC. If access cannot be gained to this resource or there are issues with the installation of the appropriate software to run HLS tools on the card, implementation of algorithms will be limited. To mitigate this risk, development of the deliverable software will be done in parallel with simulation using the Xilinx Vitas tool-chain which can be used to generate estimated results for the FPGAs performance. Due to the complexity of algorithms being developed using the simulated hardware, the computational bandwidth required is unknown. As such, the QUT HPC will be used for the execution of simulations should it be too intense for a personal computer.

Secondary risks to the project are the compatibility of portable software kernels amongst devices, constrained by the availability of devices on the QUT HPC. A goal of the project is also to investigate the portability of software written in the OpenCL language as a way to reduce development time of algorithms. However, recently AMD have deviated the OpenCL standard to the Heterogeneous-computing Interface for Portability (HIP) framework. To reduce the impact of this on the project, the code base will written in HIP and bench marked on AMD compute device to provide comparable to an OpenCL implementation.

6.2 Requirements

todo

6.3 Constraints

todo

6.4 Risk Assessment

Risk	Likelihood	Consequence	Control type	Control
Musculoskeletal strain/poor posture from prolonged sitting and computer use	Medium	Low	Administrative	Take regular breaks (15 minutes per hour), maintain correct posture, use an adjustable chair and monitor height in line to be in line with eye level with elbows at 90 degrees.
Eye strain and mental fatigue from extended screen exposure	Medium	Low	Administrative	Follow the 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds), adjust screen brightness/contrast, ensure adequate ambient lighting
Loss of thesis work, code, or data due to hardware failure, corruption, or accidental deletion	Medium	High	Engineering & Administrative	Use git version control with a remote repository, push commits to remote repository daily
Electrical or trip hazard from computer equipment, chargers, and loose cabling	Low	Medium	Engineering	Ensure all equipment is electrical tested where required, route cables away from walkways, keep the desk area tidy and free of clutter
Burnout or stress from extended, isolated independent work under project deadlines	Medium	Medium	Administrative	Set realistic milestones and a project schedule, maintain regular check-ins with supervisors, schedule rest days and enforce reasonable working hours

Table 4: Project risk assessment — should this be an appendix? (probably)

7 Quality & Sustainability

7.1 Quality

To ensure that the quality of outcome from this research, we will structure the milestone deliverables such that the codebase and corresponding simulation outputs match existing literature and verified results. This is crucial in ensuring the validity of solutions, as this project aims to work closely with novel implementations of algorithms through the use of compilation pipelines that introduce significant abstraction. This is accompanied by the staged implementation of progressively more complex biological models. By ensuring the validity of algorithms initially on simple models — Geometric Brownian Motion cite — with well understood analytical behaviour we can ascertain whether the algorithms are performing per their intention.

While validating the mathematical accuracy of models is important in ensuring the statistical usability of the research, the main focus of the work will be to explore the power usage of computation is impacted through the utilisation of reconfigurable hardware. Therefore, a key result that we will be looking to understand is the total power consumption of the device for an allotted amount of calculations. It is important to the outcomes of this research that these results are reflective of the conditions that this equipment will be used, i.e. in data centres. Hence, we will be targeting consumption readings directly from the equipment we use on the HPC. This will be accompanied by hardware simulations in preliminary findings to make initial comments on observed performance.

7.2 Sustainability

Sustainability is a key focus of this project, specifically we are exploring how reconfigurable devices in heterogeneous computing environments can lower the total power consumption for a HPC with equivalent computational performance. While this is the end goal, we are required to do significant testing on the QUT HPC. As such, it is important to consider the short-term impact this has on the environment through via power utilisation. We aim to mitigate this by ensuring all algorithms are validated through simulation testing before being executed on the HPC.

8 Timeline & Deliverables

9 Management of Project Changes

Sign off

Name	Signature	Date
Luca Lubrano		04/09/2026

A Literature Review

introduction goes here

A.1 Field Programmable Gate Arrays

similar to MXB372, add more electrical engineering / computer science stuff

A.2 Stochastic Simulation

similar to MXB372

A.3 Monte Carlo

similar to MXB372

A.4 Mixed-precision Methods

similar to MXB372

References

[1] J. L. Abkowitz, S. N. Catlin, and P. Guttorp, “Evidence that hematopoiesis may be a stochastic process in vivo,” *Nature Medicine*, vol. 2, no. 2, pp. 190–197, Feb. 1996.

[2] A. Arkin, J. Ross, and H. H. McAdams, “Stochastic kinetic analysis of developmental pathway bifurcation in phage lambda-infected Escherichia coli cells,” *Genetics*, vol. 149, no. 4, pp. 1633–1648, Aug. 1998.

[3] M. Kærn, T. C. Elston, W. J. Blake, and J. J. Collins, “Stochasticity in gene expression: From theories to phenotypes,” *Nature Reviews Genetics*, vol. 6, no. 6, pp. 451–464, Jun. 2005.

[4] D. J. Wilkinson, “Stochastic modelling for quantitative description of heterogeneous biological systems,” *Nature Reviews Genetics*, vol. 10, no. 2, pp. 122–133, Feb. 2009.

[5] P. Guptasarma, “Does replication-induced transcription regulate synthesis of the myriad low copy number proteins of Escherichia coli?” *BioEssays*, vol. 17, no. 11, pp. 987–997, 1995.

[6] M. B. Elowitz, A. J. Levine, E. D. Siggia, and P. S. Swain, “Stochastic Gene Expression in a Single Cell,” *Science*, vol. 297, no. 5584, pp. 1183–1186, Aug. 2002.

[7] A. Raj and A. van Oudenaarden, “Nature, Nurture, or Chance: Stochastic Gene Expression and Its Consequences,” *Cell*, vol. 135, no. 2, pp. 216–226, Oct. 2008.

[8] D. J. Warne, R. E. Baker, and M. J. Simpson, “Simulation and inference algorithms for stochastic biochemical reaction networks: From basic concepts to state-of-the-art,” *Journal of The Royal Society Interface*, vol. 16, no. 151, p. 20 180 943, Feb. 2019.

[9] L. Lannelongue and M. Inouye, “Carbon footprint estimation for computational research,” *Nature Reviews Methods Primers*, vol. 3, Feb. 2023.

[10] L. Lannelongue *et al.*, “GREENER principles for environmentally sustainable computational science,” *Nature Computational Science*, vol. 3, no. 6, pp. 514–521, Jun. 2023.

[11] R. Brightwell, B. W. Barrett, K. S. Hemmert, and K. D. Underwood, “Challenges for High-Performance Networking for Exascale Computing,” in *2010 Proceedings of 19th International Conference on Computer Communications and Networks*, Zurich, Switzerland: IEEE, Aug. 2010, pp. 1–6.

[12] G. Rostirolla, R. da Rosa Righi, V. F. Rodrigues, P. Velho, and E. L. Padoin, “GreenHPC: A novel framework to measure energy consumption on HPC applications,” in *2015 Sustainable Internet and ICT for Sustainability (SustainIT)*, Apr. 2015, pp. 1–8.

[13] D. J. Warne, R. Hayward, N. Kelson, J. Banks, and L. Mejias, “Pulse-coupled neural network performance for real-time identification of vegetation during forced landing,” *ANZIAM Journal*, vol. 54, p. 1, Mar. 2014.

[14] D. J. Warne, N. A. Kelson, and R. F. Hayward, “Comparison of High Level FPGA Hardware Design for Solving Tri-diagonal Linear Systems,” *Procedia Computer Science*, vol. 29, pp. 95–101, 2014.

[15] I.-B. Haas and M. B. Giles, *A nested MLMC framework for efficient simulations on FPGAs*, Feb. 2025.

[16] T. Kimpson, M. B. Flegg, and J. A. Flegg, *Reduced-Precision Stochastic Simulation For Mathematical Biology*, May 2026.

[17] C. Brugger *et al.*, “Mixed precision multilevel Monte Carlo on hybrid computing systems,” in *2014 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFEr)*, London, UK: IEEE, Mar. 2014, pp. 215–222.

[18] N. Metropolis and S. Ulam, “The Monte Carlo Method,” *Journal of the American Statistical Association*, vol. 44, no. 247, pp. 335–341, Sep. 1949.

[19] J. Dick, F. Y. Kuo, and I. H. Sloan, “High-dimensional integration: The quasi-Monte Carlo way,” *Acta Numerica*, vol. 22, pp. 133–288, May 2013.

[20] M. B. Giles, “Multilevel Monte Carlo methods,” *Acta Numerica*, vol. 24, pp. 259–328, 2015.

[21] M. B. Giles, “Multilevel Monte Carlo Path Simulation,” *Operations Research*, vol. 56, no. 3, pp. 607–617, Jun. 2008.

[22] D. J. Warne, N. A. Kelson, and R. F. Hayward, “Solving Tri-Diagonal Linear Systems using Field Programmable Gate Arrays,”

[23] J. Banks *et al.*, “CALCULATION OF ENGINE PARAMETERS USING RECONFIGURABLE HARDWARE,” *International Journal of Computational Methods*, vol. 11, no. supp01, p. 1 344 011, Nov. 2014.

[24] C. De Schryver, Ed., *FPGA Based Accelerators for Financial Applications*. Cham: Springer International Publishing, 2015.

[25] D. J. Warne, S. A. Sisson, and C. Drovandi, “Vector Operations for Accelerating Expensive Bayesian Computations – A Tutorial Guide,” *Bayesian Analysis*, vol. 17, no. 2, Jun. 2022.

[26] H. J. Macintosh, J. E. Banks, and N. A. Kelson, “Implementing and Evaluating an Heterogeneous, Scalable, Tridiagonal Linear System Solver with OpenCL to Target FPGAs, GPUs, and CPUs,” *International Journal of Reconfigurable Computing*, vol. 2019, pp. 1–13, Oct. 2019.

[27] D. J. Warne, K. Lindsay, and S. Hurn, “A Heterogeneous Computing Approach to Simulation of the Heston Stochastic Volatility Model,” *ANZIAM Journal*, vol. 57, p. 364, Nov. 2016.

[28] H. J. Macintosh, D. J. Warne, N. A. Kelson, J. E. Banks, and T. W. Farrell, “Implementation of parallel tridiagonal solvers for a heterogeneous computing environment,” *ANZIAM Journal*, vol. 56, p. 446, Feb. 2016.

[29] J. M. P. Cardoso, J. G. F. Coutinho, and P. C. Diniz, “Chapter 2 - High-performance embedded computing,” in *Embedded Computing for High Performance*, J. M. P. Cardoso, J. G. F. Coutinho, and P. C. Diniz, Eds., Boston: Morgan Kaufmann, Jan. 2017, pp. 17–56.

[30] D. T. Gillespie, “The chemical Langevin equation,” *The Journal of Chemical Physics*, vol. 113, no. 1, pp. 297–306, Jul. 2000.

[31] I.-B. Haas and M. B. Giles, *A nested MLMC framework for efficient simulations on FPGAs*, Feb. 2025.

[32] T. Jahnke and W. Huisinga, “Solving the chemical master equation for monomolecular reaction systems analytically,” *Journal of Mathematical Biology*, vol. 54, no. 1, pp. 1–26, Jan. 2007.

[33] L. Michaelis and M. L. Menten, “Die Kinetik der Invertinwirkung,” *Biochemische Zeitschrift*, vol. 49, pp. 333–369, 1913.

[34] R. Marshall and V. Noireaux, “Quantitative modeling of transcription and translation of an all-E. coli cell-free system,” *Scientific Reports*, vol. 9, no. 1, p. 11 980, Aug. 2019.

[35] M. B. Elowitz and S. Leibler, “A synthetic oscillatory network of transcriptional regulators,” *Nature*, vol. 403, no. 6767, pp. 335–338, Jan. 2000.

[36] O. Pulkkinen and R. Metzler, “Variance-corrected Michaelis-Menten equation predicts transient rates of single-enzyme reactions and response times in bacterial gene-regulation,” *Scientific Reports*, vol. 5, no. 1, p. 17 820, Dec. 2015.

[37] F. Wong, A. Dutta, D. Chowdhury, and J. Gunawardena, “Structural conditions on complex networks for the Michaelis–Menten input–output response,” *Proceedings of the National Academy of Sciences*, vol. 115, no. 39, pp. 9738–9743, Sep. 2018.